

PROBLEM STATEMENT 26090

Implementation Plan

AI-Driven Market Linkage & Smart Cataloging Application

A production-grade AI-powered Virtual Business Manager for marginalized artisans, weavers, and micro-entrepreneurs — built for the Ministry of Social Justice and Empowerment (MoSJE).

Capture

Speak

Sell

1. Background & Problem Context

The Ministry of Social Justice and Empowerment (MoSJE) provides financial assistance to artisans, weavers, and micro-entrepreneurs from marginalized communities. Physical melas (Shilp Samagam, Surajkund Mela, Dilli Haat) create temporary sales spikes, but artisans lack a continuous, year-round digital sales channel. This project builds that channel.

Critical Pain Points

- **Low Digital Literacy & Language Barrier** — traditional e-commerce seller dashboards are overwhelming and complex for this audience.
- **Photography & Presentation Gap** — raw smartphone photos look unprofessional next to studio-shot e-commerce products.
- **Cataloging & Description Barrier** — difficulty writing SEO-friendly English/Hindi titles, descriptions, and craft specifications.
- **Suboptimal Pricing** — no visibility into market trends leads to underpricing or unsellable overpricing.

Core Solution

An intuitive, accessible, **voice-first, mobile-responsive web application** — the “Virtual Business Manager” — that reduces product digitization to a simple three-step workflow: **Capture** → **Speak** → **Sell**.

2. System Architecture Overview

The application is organized into four functional pillars that sit on top of a shared backend and data layer. Each pillar maps directly to one stage of the artisan's workflow, and together they feed a central admin analytics layer for MoSJE.

Pillar	Purpose
1. AI Studio	Turns raw product photos into clean, professional e-commerce images.
2. Voice Cataloger	Converts spoken product descriptions (regional language) into structured, SEO-ready listings.
3. Dynamic Pricing	Recommends a fair, market-aware price range from cost and labor inputs.
4. Smart Inventory & Linkage	Publishes the finished listing to buyers via storefront, PDF/WhatsApp catalog, B2B inquiries, and ONDC/GeM.

All four pillars read from and write to a shared data layer (artisan profiles, product listings, pricing history, B2B inquiries), which the MoSJE Admin Dashboard aggregates for impact reporting.

3. Component-by-Component Breakdown

COMPONENT 1

AI Image Enhancer & Studio

Turns a messy smartphone photo into a clean, sellable product image — no photography skill required.

- **Interactive Camera/Upload** — drag-and-drop or camera capture with on-screen framing guides.
- **AI Background Removal** — client-side edge-detection and canvas masking to strip cluttered backgrounds.
- **Studio Backdrop Presets** — Clean White, Warm Marble, Wooden Surface, Fabric Texture, Soft Shadow.
- **Auto Lighting & Sharpness Correction** — canvas filters for contrast, brightness, saturation, and drop-shadow generation.

COMPONENT 2

Multilingual Auto-Cataloger (Voice-to-Listing)

Lets an artisan simply speak about their product in their own language — the system writes the listing.

- **Voice Note Recorder** — large mic button, records in Hindi, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, or Hinglish.
- **Speech Processing & NLP Engine** — speech-to-text transcription with auto-translation into professional English and Hindi.
- **Smart Catalog Generator** — auto-fills SEO product title, craft type & material (e.g. Chanderi Silk, Terracotta Pottery, Dhokra Craft, Wooden Carving), key features & dimensions, washing/care instructions, and search tags/keywords.

COMPONENT 3

Dynamic ML Pricing Assistant

Stops artisans from underselling their work by giving a data-backed price range.

- **Cost Factor Inputs** — raw material cost (■), labor hours spent, craft complexity rating.
- **AI Price Calculator** — computes Minimum Base Cost (material + fair hourly wage), a Benchmark E-Commerce Range (modelled market averages for similar craft categories), and a Recommended Retail Price with a margin breakdown chart.

COMPONENT 4

Artisan Storefront & B2B Market Linkage

Turns a finished listing into something buyers — both retail and bulk — can actually find and act on.

- **Artisan Micro-Store** — a digital storefront per artisan with a shareable link and QR code.
- **Instant PDF/WhatsApp Catalog Generator** — one-click printable PDF catalog for buyers.
- **B2B Bulk Inquiry Portal** — direct inquiry form for wholesale buyers, craft stores, and government procurement officers.
- **ONDC / GeM Schema Exporter** — exports products as JSON structured for the Open Network for Digital Commerce and Government e-Marketplace.

COMPONENT 5

MoSJE Admin & Impact Analytics Dashboard

Gives the Ministry a live view of reach and economic impact across all onboarded artisans.

- **Artisan Onboarding Metrics** — total artisans registered, active product count, regional breakdown.
- **Economic Upliftment Tracking** — estimated revenue generated, average listing price, B2B inquiries matched.
- **Multilingual Support** — live language switcher across Hindi, English, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada.

4. UI/UX Design System for Low-Literacy Accessibility

Because the primary users may have limited digital literacy, the interface leans on visual and audio cues rather than dense text.

Visual Hierarchy

- Icon-heavy buttons, high-contrast colors, and audio feedback hints throughout.

Color Palette

Swatch	Color	Role
	Deep Royal Indigo — #1E1B4B	Primary — traditional Indian heritage
	Terracotta Warm Amber — #D97706	Primary accent — traditional Indian heritage
	Emerald Green — #059669	Positive pricing & sales metrics

Theming & Typography

- Dark/light theme support with subtle glassmorphism touches.
- Typography: Google Fonts Inter / Outfit, chosen for ultra-clean readability across screen sizes.

5. Verification & Testing Plan

Area	What to Verify
AI Background Removal	Upload cluttered craft images and confirm background-removal precision and correct studio-background insertion.
Voice Cataloger	Test voice input across multiple simulated regional-language descriptions; confirm accurate auto-generation of titles, descriptions, and SEO tags in English & Hindi.
Pricing Engine	Run dynamic pricing calculations across craft types (Textiles, Pottery, Metalwork, Woodwork) with varying material costs and labor hours.
Export & Sharing	Test PDF catalog generation, QR code rendering, and ONDC/GeM schema output.
Responsive & Multilingual UX	Test language switching on every screen and confirm UI responsiveness on mobile and tablet viewports.